

User Manual – AMS Display V2.3

The AMS Display allows convenient management of filament spools directly on the device, via the web interface or with the optional AMS Dial.

Display Operation

◆ Overview

- Displays all AMS slots (4 or 8)
 - Each tile represents one slot
 - Colors indicate material and status
 - Manage up to 32 filament spools
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Touch Functions

◆ Short tap on a slot

Opens the NumPad to subtract used filament.

- Enter the desired weight in grams
 - Press "OK" to confirm
 - The weight is subtracted from the spool
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◆ Long press on a slot

Opens the spool detail view.

Displays:

- Spool number
- Material / Type
- Color
- Manufacturer
- Weight
- Article number

- Timestamp of the last modification
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- ♦ Inside the spool popup
 - ❌ X → closes the window
 - 🔄 Swap → opens the Swap NumPad
 - 📶 Write NFC → writes the current spool data to an NFC tag
 - 📶 Read NFC → reads an NFC tag and synchronizes its data with the display
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🔄 Swapping Spools

1. Select a spool (long press)
2. Press "Swap"
3. Enter the target spool number
4. Press "OK"

👉 The two spools will be swapped.

1 2 3 4 NumPad Functions

- Numbers → input
 - DEL → deletes the last digit
 - OK → confirms
 - X → cancels
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⚙️ Configuration Page

The configuration page provides several system settings.

- ♦ AMS Mode
- 4 Spools
- 8 Spools

The display layout is updated immediately.

◆ Web UI

- ON → Web interface enabled
 - OFF → Web interface disabled
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◆ MQTT

- ON → MQTT enabled
- OFF → HTTP mode

MQTT is completely optional.

The display works fully without MQTT.

◆ Display Rotation

Supports two installation orientations:

- USB Port TOP
- USB Port BOTTOM

Display rotation and touch alignment are adjusted automatically.

The setting is stored permanently.

◆ Language

- German
- English

The language can be changed at any time.

The setting is stored permanently.

 Web UI Operation

The web interface can be accessed through the device IP address.

◆ Overview

- Displays all spools

- Filter options:

☐ All

☐ AMS

☐ Storage

◆ Spool Cards

- 👉 Each card provides three main functions:

- Tap on the card

→ Opens the weight input popup

- "Swap" button

→ Starts the spool swap process

- "Edit" button

→ Opens the spool editing window

◆ Edit Spool

The following data can be modified:

- Material

- Type

- Color

- Article number

- Manufacturer

- Empty status
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◆ Empty Function

- Enable the "Empty" checkbox
- The spool will be marked as empty

Display:

Empty (Spool X)

◆ Import / Export

- 📁 Export → saves all spool data as a JSON file
- 📁 Import → loads a previously saved JSON file

The exported file can be used as a backup or transferred to another display.

◆ WiFi Setup

- Enter SSID and password
 - Enter Display IP address
 - Enter MQTT IP address (optional)
 - Save the settings
 - The device will restart automatically
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◆ Factory Reset

The reset button deletes:

- WiFi settings
- Display IP address
- MQTT IP address

After reboot the display automatically starts in AP mode again.

📶 WiFi Behavior

- First startup: Access Point mode
 - After configuration: Connects to your WiFi
 - On connection failure: Returns to AP mode
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WiFi Status Display

-  Connected → WiFi + IP address
 -  AP Mode → Setup network active
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AP Setup Mode

SSID: AMS-Setup

Password: 12345678

Open in your browser:

<http://192.168.4.1>

MQTT Functions (Optional)

When MQTT is enabled:

- Transfer of:
 - ☐ Spool data
 - ☐ Weight information
 - ☐ Status messages
 - ☐ AMS Dial commands
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Possible Integrations

- Automatic monitoring
- Filament usage logging
- Smart Home integration

- Home Assistant
 - Node-RED
 - Custom systems
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HTTP Fallback

If no MQTT server is available, the display automatically switches to HTTP mode.

All functions remain fully operational without MQTT.

AMS Dial (Optional)

The AMS Dial allows:

- Selecting a spool
- Changing filament weight
- Swapping spools

Communication works automatically via MQTT or HTTP.

NFC Functions

The display supports NFC tags using a PN532 reader.

Stored data:

- Spool number
- Remaining weight
- Material
- Type
- Color
- Manufacturer
- Article number
- Empty status

- Timestamp
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NFC Synchronization

When reading an NFC tag, timestamps are compared automatically.

- NFC is newer → data is imported into the display.
- Display is newer → the NFC tag is updated.
- Both are identical → no action is taken.

This makes it easy to synchronize filament spools between multiple printers or even different users.

Notes

- Slots are independent from spools.
 - Spools can be swapped freely.
 - All data is stored permanently.
 - Settings remain saved after reboot.
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Tips

- Always confirm weight changes with OK.
 - Enter the correct spool number when swapping.
 - Check WiFi credentials carefully.
 - Do not remove NFC tags until the write process has finished.
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Support

For questions or issues:

- Use the GitHub repository
- Feedback is always welcome.